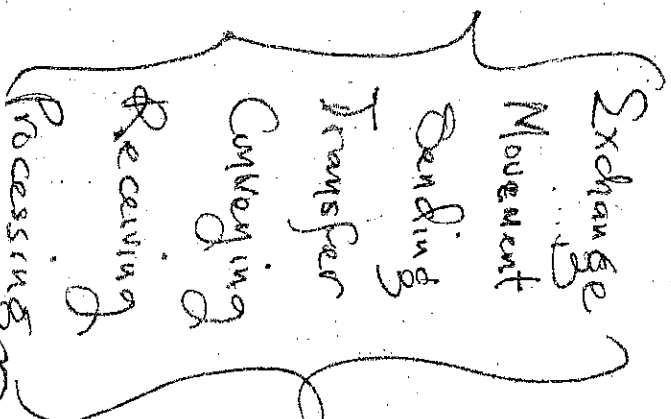


HND 1 222 325 (Winter 2021)

Communication System;

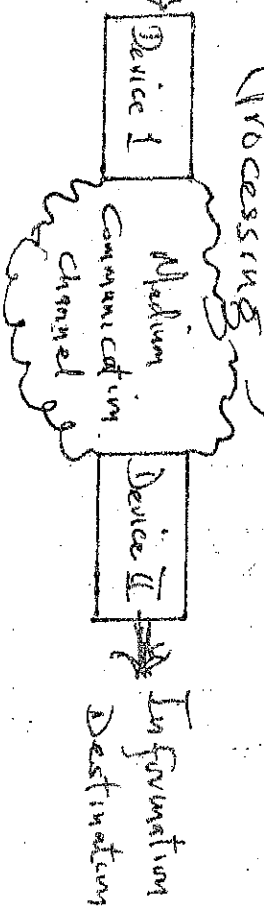
Basically Telecommunication Engineering is concerned with the transmission of information between two distant points.

Information —

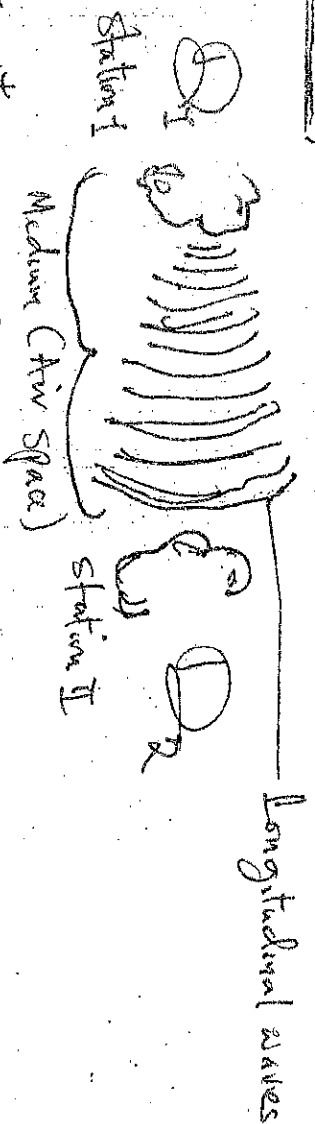


Functions between stations,
Using acceptable Medium.

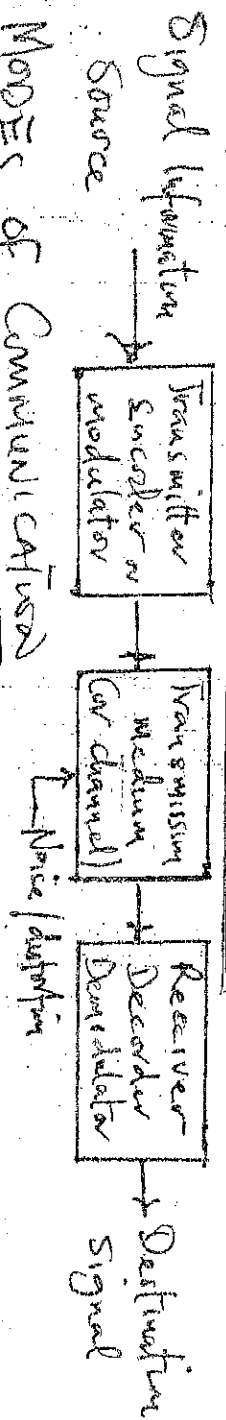
Information Source;



Analogous Source;



BASIC ELEMENTS OF A TELECOMM. SYSTEM;



MODES OF COMMUNICATIONS

- Simplex ————— $D_1 \rightarrow D_2$ one way
- Half Duplex ————— $D_1 \leftarrow D_2$ one way at a time
- Full Duplex ————— $D_1 \rightarrow D_2$ Two ways Simultaneously

Communication System Basics

Information Source $\rightarrow$ A signal Carrying Intelligence Signal = Fundamental quantity which is time variant (function of time).

Physical Quantities - (Mass, Pressure, Sound, voltage, current) all vary with time = Signals - Forms.

For Digital Electronics = Voltage | Current as functions of time

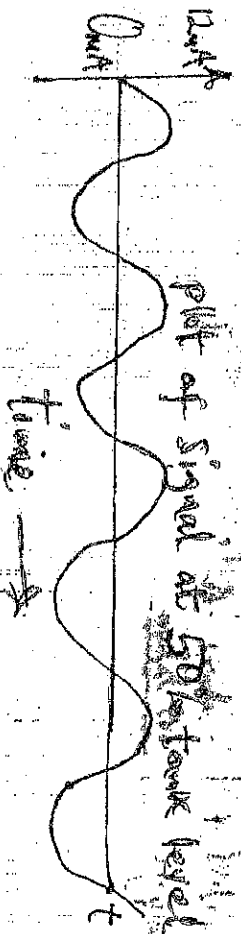
BASIC TYPES OF SIGNALS

There are various types of signals in electronics & telecommunication engineering, but in general, they can be broadly classified into two major categories, these includes;

- Continuous-time signal
- Discrete-time signal

1. CONTINUOUS-TIME SIGNAL;

These are also called Analogue time signals which has any value between a given limit, when the independent variable is Continuous.



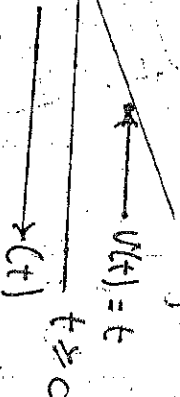
The above diagram is a simple illustration of analogue signal (continuous-time). Any electrical noise that is introduced into an analog measurement system will be interpreted as change in the measure quantity.

Examples of Continuous time Signals

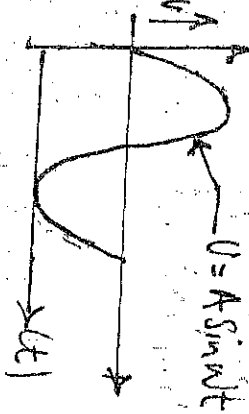
I - Constant (static) Signal $\rightarrow V \uparrow$



II - Ramp Signal $\rightarrow V \uparrow$



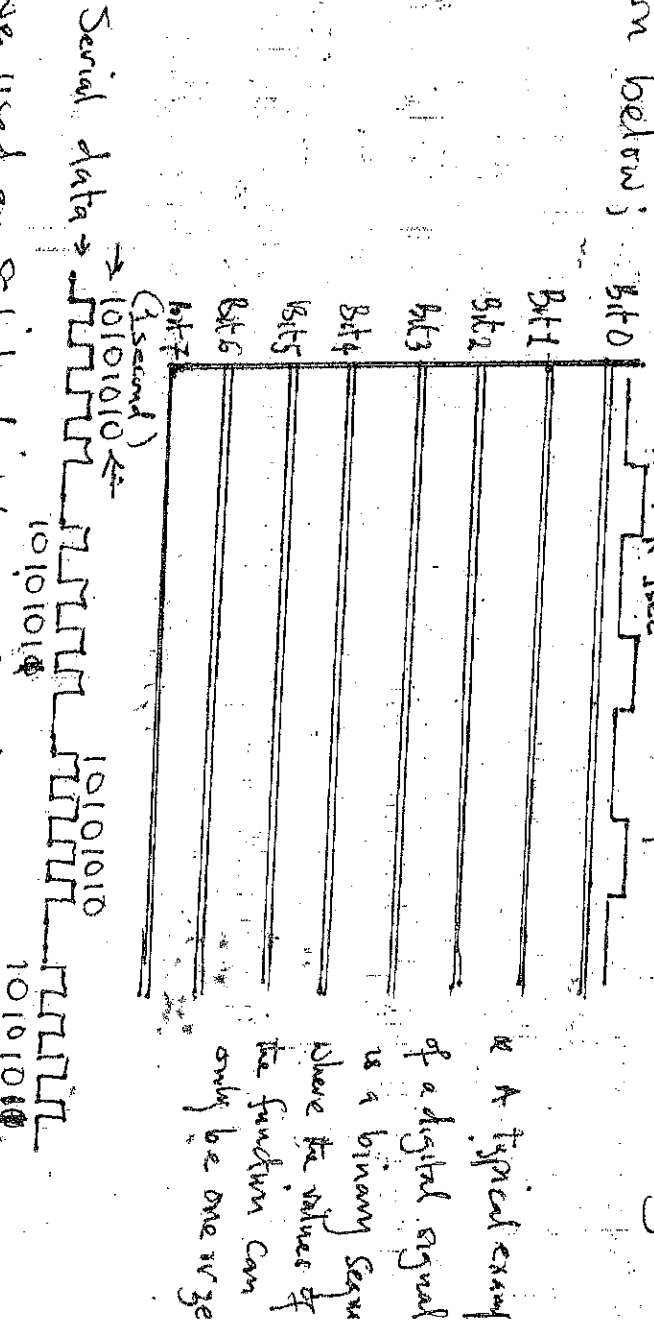
III - Sinusoidal Signal



It is important to note that, our speeches, the way we listen are all analogue, in fact the real world is analogue.

Discrete-Time-Signal:

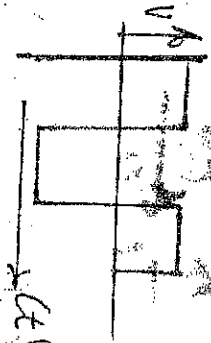
This is also called Digital Signal, they are signals that have fixed number of possible values, when the independent variables takes a discrete value. Simple illustration of discrete-time-signal, shown below:



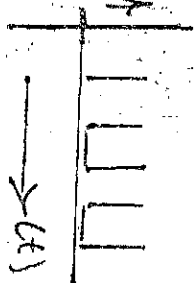
If we used an 8-bit digital signal serial network, all data bits must appear on the single channel in sequence. And these bits must be output to the transmitter within the 1-second window time between A/D Converter update. Therefore, the alternating digital output of 10101010 and 10101011 (once per second) would be shown as illustrated above.

Examples of Discrete time - Signals;

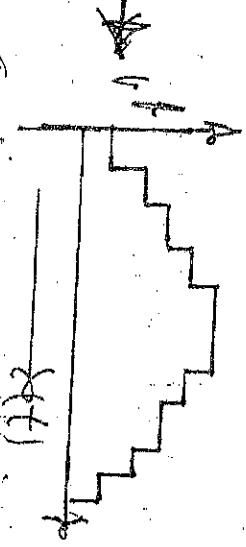
i) Rectangular Pulse Signal —



ii) Impulse Signal —



iii) Step - Signal



Quantized Discrete signal

Other classifications include = Real Signals i.e. FM, AM, VLF signals.

Audio, Telephone, TV signals.

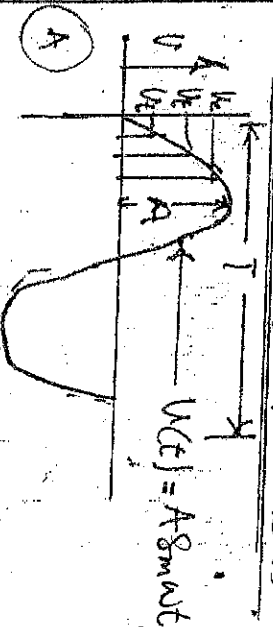
PROPERTIES OF SIGNAL

i) Periodic in nature — This measure the frequency of repetition of the signal

ii) Distortion — Measure of deviation from original nature due to noise or external interference.

iii) Frequency / Phase - Shift - Deviation due to modulation.

SIGNAL PARAMETERS

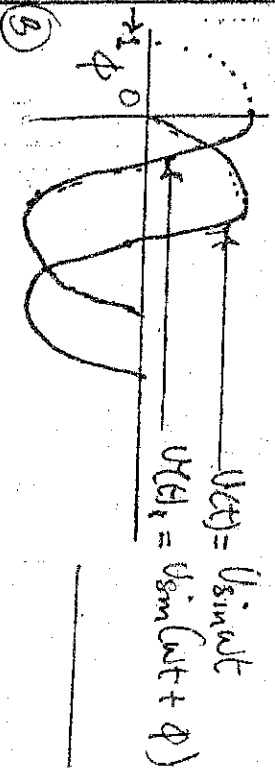


where $A = \text{Amplitude}$

$\omega = \text{Angular velocity in RPS}$

$V(t) = \text{Instantaneous value of } V$

$$\therefore \omega = 2\pi f$$



where $\phi = \text{phase angle}$

* $\phi = \text{Phase diff. at } t=0$

* $P_t = \text{Amplitude of the integral of the}$

Modulation And Its Needs

5

This is the process where by an intelligent signal (such as speech) of low frequency is superimposed on a carrier wave of higher frequency for the purpose of transmission.

Need for Modulation:

It has been observed that before energy of signal is radiated from an antenna, the frequency of the signal must be above 10 KHz. Meaning that, the more the frequency of operation the more the radiation. Since the intelligent signal is normally of low frequency, it has to be superimposed the higher frequency carrier (sinusoidal wave) or modulating signal.

- Other reason is to reduce the size of the tuned antenna,
- Efficiency in transmission
- channel diversification

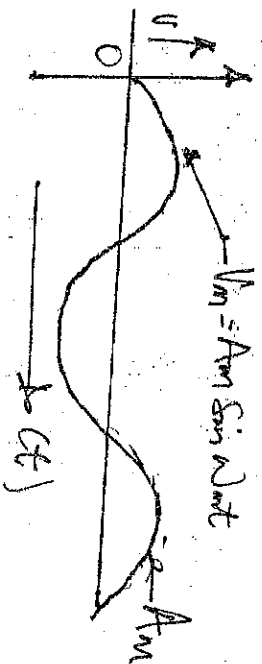
Types of Modulation

- Amplitude modulation (AM)
- Angle modulation $\rightarrow$ frequency modulation (FM)
 $\rightarrow$ phase modulation (PM)

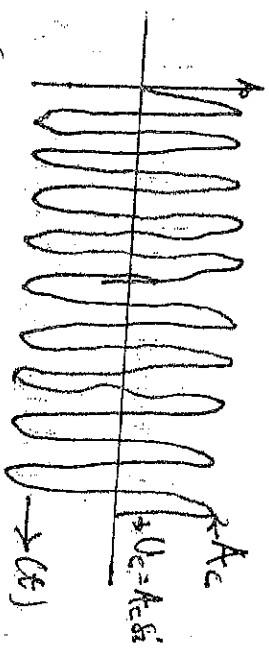
Amplitude Modulation (AM)

This is a modulation technique where the amplitude of a carrier wave is varied by another modulating voltage (ie. information signal) whose frequency is invariably lower than that of the carrier wave. The amplitude of the carrier is made proportional to the instantaneous amplitude of the

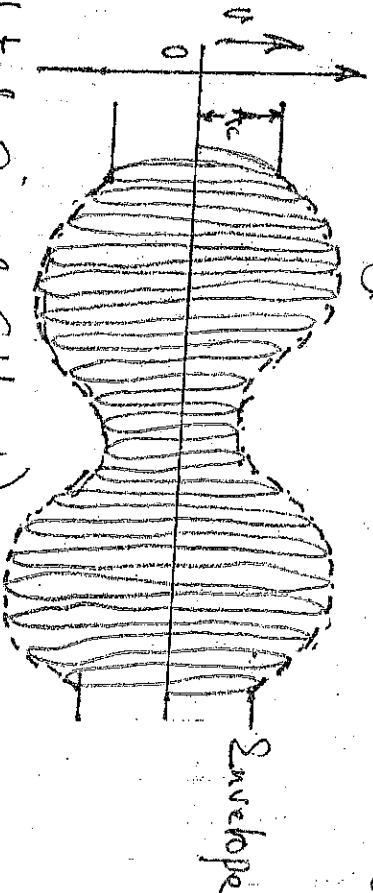
Modulating Signal, giving birth to a resultant modulate wave which has three components; the original carrier wave and two other signals of different frequencies called the Sidebands.



② Sinusoidal Modulating wave ($V_m(t)$)



③ Sinusoidal Carrier wave



④ Modulated Signal $V_c(t)$

Where V_c = the instantaneous value of the Carrier sine wave voltage

A_c = the Peak value of the Constant unmodulated Carrier

Since wave is measured b/w zero and the Minimum amplitude in both directions. (Reference point or Peak for the modulating signal)

ω_c = the Angular velocity measured in radians also given by $2\pi f_c$ where f_c is the freq. of the Carrier sine wave

t_1 = is a particular Point in time during the Carrier Cycle

Note that, the modulated sine wave carries an imaginary line connecting the positive and negative peaks of the carrier wave form which is known as the ENVELOPE.

In general, it is expected that the amplitude of the modulation signal be less than that of the carrier wave to avoid distortion.

If the general formula for a sinusoidal carrier wave is given by;

$$V_c(t) = A_c \cos(2\pi f_c t + \phi) \quad \text{--- (1)}$$

where A_c , f_c and ϕ may be varied for the purpose of transmitting information giving an amplitude, frequency and phase modulation respectively.

Then, let $A = K + V_m(t)$

where K is the unmodulated carrier amplitude and $V_m(t) = a \cos(2\pi f_m t)$ is the modulating signal.

The modulated carrier assumed to be represented as;
 $\Rightarrow \phi = 0$ is $V_c(t) = [K + a \cos(2\pi f_m t)] \cos(2\pi f_c t)$
 that is;

$$V_c(t) = K [1 + M \cos(2\pi f_m t)] \cos(2\pi f_c t) \quad \text{--- (2)}$$

where M is the depth of modulation and it is defined

$$M = \frac{\text{modulating signal amplitude}}{\text{unmodulated carrier amplitude}} = a/K$$

Also $A_m < A_c$ and the ratio A_m/A_c is known as the modulation factor, modulation index or coefficient of modulation

$$\therefore M = A_m/A_c$$

where $0 < M \leq 1$ - for an undistorted envelope

Therefore, $\therefore$ If this condition is not observed the envelope becomes distorted and a carrier phase reversal occurs, as shown above in eqn. II.

Equation I - may be expanded to yield;

$$V_c(t) = K [\cos(2\pi f_c t) + \frac{M}{2} \cos(2\pi(f_c - f_m)t) + \frac{M}{2} \cos(2\pi(f_c + f_m)t)]$$

* Modulation index is defined as the measure of extent of amplitude variation about $\sim 1.1110 \dots$

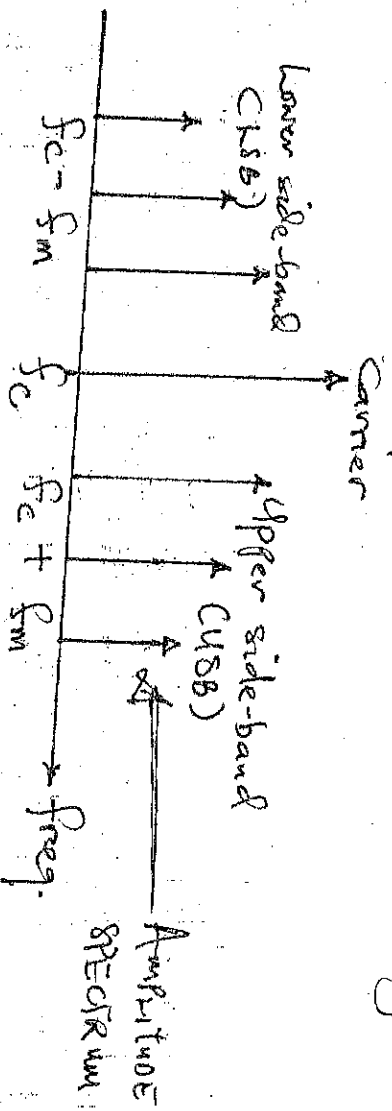
For Amplitude Spectrum of the modulated carrier,
Therefore,

$$V_{m(t)} = a_1 \cos(2\pi f_1 t) + a_2 \cos(2\pi f_2 t) + a_3 \cos(2\pi f_3 t) + \dots$$

The modulated carrier becomes,

$$V_{c(t)} = K \left[1 + M_1 \cos(2\pi f_1 t) + M_2 \cos(2\pi f_2 t) + M_3 \cos(2\pi f_3 t) + \dots \right] \cos 2\pi f_c t$$

It is seen that the amplitude spectrum of a modulated carrier basically consist of three components, that is, The carrier frequency (f_c), the lower side frequency or the upper side frequency, ($f_c - f_m$) and ($f_c + f_m$) this is called DOUBLE-SIDE BAND AMPLITUDE MODULATION. And the amplitude spectrum of (DSB)-AM is given as;



Note that, A circuit that produced AM is called a Modulator. * A Circuit that compute the product of two analog signals are known as analog multipliers, mixers, converters or detectors. A circuit which changes lower freq. baseband to a higher freq. signal is usually called a MODULATOR.

Example;

A Tuned Circuit of the Oscillator in a Simple AM transmitter employs a 50 microhenry (50 μ H) coil and a 1-nanofarad (1 nF) capacitor. If the oscillator output is modulated by audio freq. up to 10KHz, what will be the freq range occupied by the sideband?

$$f_c = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{5 \times 10^{-5} \times 1 \times 10^{-9}}}$$

$$\frac{1}{2\pi\sqrt{(5 \times 10^{-14})}}$$

$$= 7.12 \times 10^5$$

$$\approx 712 \text{ KHz}$$

$$\therefore \text{LSB} = f_c - f_m = 712 - 10 = 702 \text{ KHz}$$

$$1100 - 10 = 1090 \text{ KHz}$$

FREQUENCY MODULATION (FM)

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This is a modulation technique where the modulating signal is superimposed on the carrier wave, and the amplitude of the carrier is kept constant while its frequency is varied by the instantaneous value of the modulating signal. Like the amplitude modulation, the modulated wave produced by frequency modulation is simply generating a range of other signal waves called the sidebands but unlike the FM, modulated wave the FM sidebands range in number according to a measure called the "DEVIATION RATIO".

The Deviation Ratio is simply defined by the shift in the carrier frequency from its resting point compared to the amplitude of the modulation voltage. Mathematically expressed as

$$DR = \frac{f_{dev}}{f_{mf}}$$

The deviation ratio is basically used by regulating agencies to limit bandwidth transmission on wideband FM. The maximum deviation ratio of 5 corresponding to 8 sidebands this is the maximum ratio allowed for commercial broadcast. The greatest advantage of the FM is that the amplitude of the frequency modulated wave always remain constant at all times.

ANGLE MODULATION

Angle Modulation is an alternative to amplitude modulation and as its name implies information is transmitted by varying the phase angle of a sinusoidal signal, And it can be conveniently written as

$$\Rightarrow V_c \cos(\omega_c t + \theta(t))$$

where θ is the phase angle that is, its made proportional to the function of the modulating signal

* Phase Modulation is a technique where the amplitude of the modulating signal is used to change the carrier phase from the value it would have gotten if unmodulated.

If the Carrier is given by the expression;

$$\Rightarrow V_c(t) = A \cos \theta(t) \quad \text{--- (1)}$$

and $\theta(t) = 2\pi f_c t + \phi + 2\pi K_2 \int_0^t V_m(t) dt$

$$\therefore V_c(t) = A \cos \left[2\pi f_c t + 2\pi K_2 \int_0^t V_m(t) dt \right] \text{ --- Modulated Carrier in } f_m$$

But;

$$V_m(t) = a \cos(2\pi f_m t)$$

$$V_c(t) = A \cos \left[2\pi f_c t + 2\pi K_2 \int_0^t a \cos(2\pi f_m t) dt \right]$$

$$= A \cos \left[2\pi f_c t + \frac{K_2 a}{f_m} \sin(2\pi f_m t) \right]$$

can be represented as;

$$\Rightarrow A \cos \left[2\pi f_c t + \beta \sin(2\pi f_m t) \right] \quad \text{--- (II)}$$

where β is the modulated index

Expanding equation (I) will give;

$$V_c(t) = A \cos(2\pi f_c t) \cos[\beta \sin(2\pi f_m t)] - A \sin(2\pi f_c t) \sin[\beta \sin(2\pi f_m t)]$$

For series expansion of a trigonometric function;

$$\cos[\beta \sin(2\pi f_m t)] \text{ and } \sin[\beta \sin(2\pi f_m t)]$$

Two type of FM will be ~~image~~ ^{improve} ~~or~~ Narrow-band FM and Wide-band FM.

Therefore;

$$\cos[\beta \sin(2\pi f_m t)] = 1 - \frac{\beta^2 \sin^2(2\pi f_m t)}{2!} + \frac{\beta^4 \sin^4(2\pi f_m t)}{4!} - \dots$$

$$+ \frac{\beta^6 \sin^6(2\pi f_m t)}{6!} - \dots$$

$$\therefore \sin[\beta \sin(2\pi f_m t)] = \beta \sin(2\pi f_m t) - \frac{\beta^3 \sin^3(2\pi f_m t)}{3!} + \frac{\beta^5 \sin^5(2\pi f_m t)}{5!} - \dots$$

F.A NARROW BAND

Narrow-band FM is produced when $\beta \ll 1$ the series expansion for this case may be approximated by;

$$\cos[\beta \sin(2\pi f_m t)] \approx 1 \text{ and } \sin[\beta \sin(2\pi f_m t)] \approx \beta \sin(2\pi f_m t)$$

Therefore,

Approximately, ~~that in (2)~~ can be rewritten as;

$$V_c(t) = A \left[\cos(2\pi f_c t) - \frac{\beta}{2} \cos(2\pi(f_c + f_m)t) + \frac{\beta}{2} \cos(2\pi(f_c - f_m)t) \right]$$

CONCLUSION CASE 1

The above expression shows that narrow band is equivalent to DSB-AM with a phase shift of 180° as the lower side band. The band width in this case is identical to the band width of a DSB-AM.

It is an evident that as the value of β increases more terms in the series expansion of $\sin$ indicate.

$$\cos[\beta \sin(2\pi f_m t)] \text{ and } \sin[\beta \sin(2\pi f_m t)]$$

It becomes significant and can not be ignored.

Example; If β is varied to 0.5, the new approximation becomes $\Rightarrow$

$$\cos[\beta \sin(2\pi f_m t)] \approx 1 - \frac{1}{2}\beta^2 \sin^2(2\pi f_m t)$$

$$\Rightarrow 1 - \frac{1}{4}\beta^2 [1 - \cos(4\pi f_m t)]$$

And

$$\sin[\beta \sin(2\pi f_m t)] \approx \beta \sin(2\pi f_m t)$$

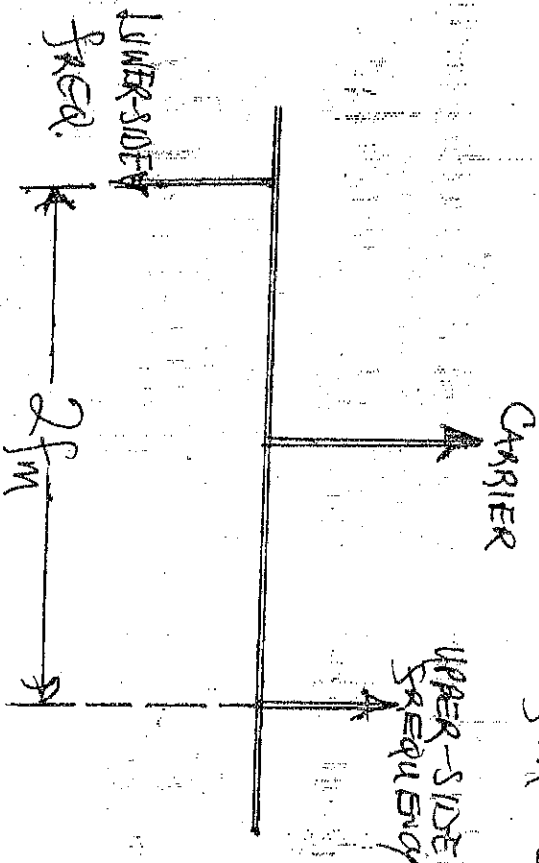
Therefore,

Equation II, can be rewrite as,

$$V_c(t) = A \left[1 - \frac{1}{4}\beta^2 \right] \left\{ \cos(2\pi f_c t) - \frac{1}{2}\beta^2 \left[\cos^2(2\pi(f_c - f_m)t) + \cos^2(2\pi(f_c + f_m)t) \right] \right. \\ \left. + \frac{1}{8}\beta^2 \left[\cos(2\pi(f_c - 2f_m)t) + \cos(2\pi(f_c + 2f_m)t) \right] \right\}$$

CONCLUSION CASE II,

The above equation resolve into two additional frequency component, spaced at $\pm 2f_m$ relative to carrier frequency. The bandwidth thus effectively doubles when compared to DSB and narrow band FM as shown below,



FREQUENCY MODULATION Vs AMPLITUDE MODULATION

- F.M Has better noise immunity when compared to A.M.
- F.M has inherent ability to pick-up the strongest signal and reject the rest making it resistant to coherent interference. And hence excellent sound production quality as compared to A.M.
- F.M requires wider frequency band in transmitting media than that of A.M.
- Finally F.M transmitters and receivers are more complex in nature than that of A.M.

* The term modulate means regulate, which the process of regulating is modulation.

NUMERICAL EXAMPLE OF F.M

A sinusoidal modulating signal given by,

$M(t) = 4 \cos 2\pi 4 \times 10^3 (t)$ is applied to an F.M modulator that has a frequency deviation constant gain of 10 kHz

Compute (a) the peak frequency deviation and

b) The modulation index. * Δf Peak deviation of the instantaneous freq. (Hz) from the centre carrier freq.

- Frequency deviation constant is $K_f = 10 \text{ kHz/V}$
- The modulating frequency $f_m = 4 \text{ kHz}$.

(a) Peak deviation occurs when the instantaneous phase of the input signal is at its maximum given by $M(t)$
The maximum value = 4 Volt . Then the $\Delta f = 4 \times 10 \text{ kHz/V} = 40 \text{ kHz}$

$$b) \beta = \frac{K_a}{f_m} = \frac{\Delta f}{f_m}$$

$$\therefore \frac{40}{4} = 10\%$$

* Noise Quieting

The noise power decreases as the signal power increases, then the SNR goes up significantly.

WIDE-BAND FM

If $\beta \gg 1$, therefore the modulation is known as wide-band FM and its bandwidth is approximately $2f_m \Delta$. And also wide-band FM uses more bandwidth. It can improve signal-to-noise ratio significantly.

With a tone-modulated FM wave, if the modulation frequency is held constant and the modulation index is increased, the (non-negligible) bandwidth of the FM signal increases but the spacing between spectra stays the same; some spectral components decrease in strength as others increase. If the frequency deviation is held constant and the modulation frequency increased, the spacing between spectra will also be increased.

CARSON'S RULE (Rule of Thumb)

It states that nearly all (98%) of the power of a frequency modulated signal lies within a bandwidth B_T .

$$\Rightarrow B_T = 2(\Delta f + f_m)$$

where Δf is the peak deviation of the instantaneous frequency from the carrier frequency.

FOURIER SERIES

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Four of the most important theorems in the theory of Fourier analysis are the inversion theorem, the Cauchy theorem, the differentiation theorem, and Parseval's theorem. All of these are based on the orthogonality of the basis functions of the Fourier series and integrals, and all require knowledge of the convergence of the sums and integrals. The practical and theoretical use of Fourier analysis is greatly expanded if use is made of distributions. Generalized functions. Because energy is an important measure of a function in signal processing, the Hilbert space of L^2 functions is a proper setting for the basic theory and a geometric view can be very useful.

Fourier Series is the series that is used to express periodic time function in the frequency domain as a function which have amplitude and repeat itself during a specific time period T .

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos(n\omega t) + b_n \sin(n\omega t) \right]$$

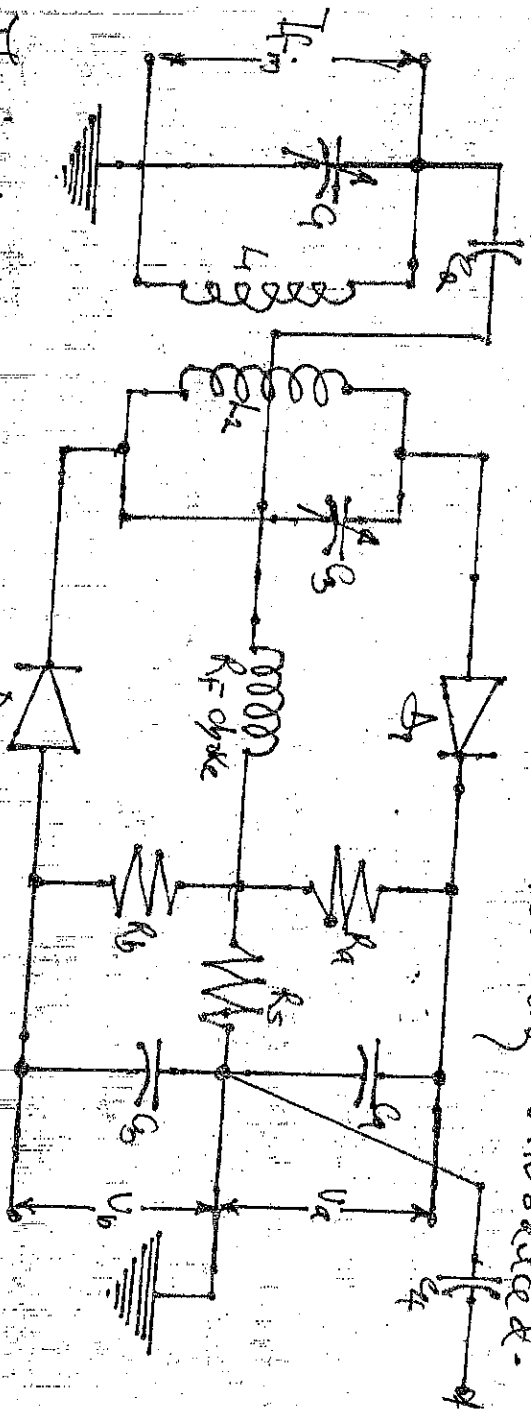
where $\frac{1}{T} = f_0$ - The fundamental frequency

n = Harmonic Number & it takes on values of $1, 2, 3, \dots$

RATIO DETECTOR

(83)

In the Foster-seely Discriminator, changes in the magnitude of the input signal will give rise to amplitude changes in the resulting output voltage. It makes prior limiting necessary, leading to possible modifying the discriminator circuit in order to provide and improve limiting function. Therefore one diode is reverse biased out, so that the amplitude limiter in the same, but the deviation from this ideal does not result in undue distortion in the ratio detector enough. Some distortion is undoubtedly introduced.



It is noted that three (3) important changes have been made; 1) One of the diodes has been reversed.

2) A large capacitor (C_4) has been placed across, used as the output, and the output now is taken from elsewhere.

3) They are also widely used in practice especially in TV receivers for the sound section and sometimes in narrow band FM radio receivers. Its basic advantages is that it provides both limiting and ratio detection.

$\sum_{n=1}^N =$ The sum of n terms, where n takes on values of 1, 2, 3, 4, ...
 $a_0, a_n, b_n =$ Amplitude or peak values of the trigonometric terms (The Fourier coefficients)

$T =$ Period of the function (waves)

$f(t) = A_n$ indicates that the Fourier series is a function of n . Only Cosine term (even) or Sine term (odd)

Therefore,

From equation 1, will derive the first ~~four~~ term

$$f(t) = \frac{a_0}{2} + \left[a_1 \cos 2\pi f t + b_1 \sin 2\pi f t \right] + \left[a_2 \cos 4\pi f t + b_2 \sin 4\pi f t \right] + \left[a_3 \cos 6\pi f t + b_3 \sin 6\pi f t \right] + \dots$$

Why Study Fourier Series In Digital Comm

It is very important to know the bandwidth occupied by the modulating signal itself. This will be useful

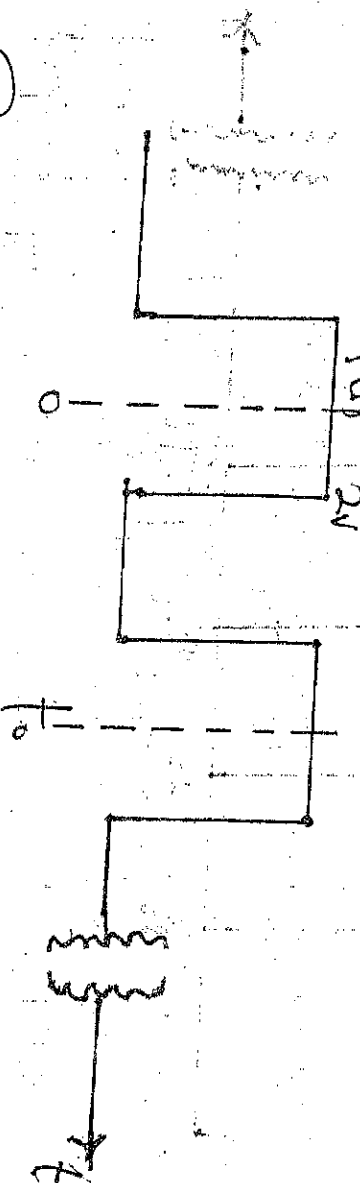
- 1.) Selection of appropriate cable for transmission.
- 2.) Selection of filters to remove unwanted signals.
- 3.) Allocation of channels to prevent co-channel interference.

Now, if the modulating signal consists of sinusoidal signals, it means that there will be no problem as the bandwidth will be the frequency range between the low and highest sine wave signal since a pure sine wave contains no harmonics.

But if the modulating signal is "Non-Sinusoidal", a more complex situation will eventually result. Since such a non-sinusoidal wave (pulses) occur frequently as modulation signal in communication, their frequency requirement is discussed as below;

FOURIER ANALYSIS OF PULSES

For a given pulses below as;



From equation (1) the above Non-sinusoidal wave can be completed using same Fourier series as shown below;

$$a_0 = \frac{1}{T_0} \int_{T_0} f(t) dt$$

$$a_n = \frac{2}{T_0} \int_{T_0} f(t) \cos(2\pi n f_0 t) dt$$

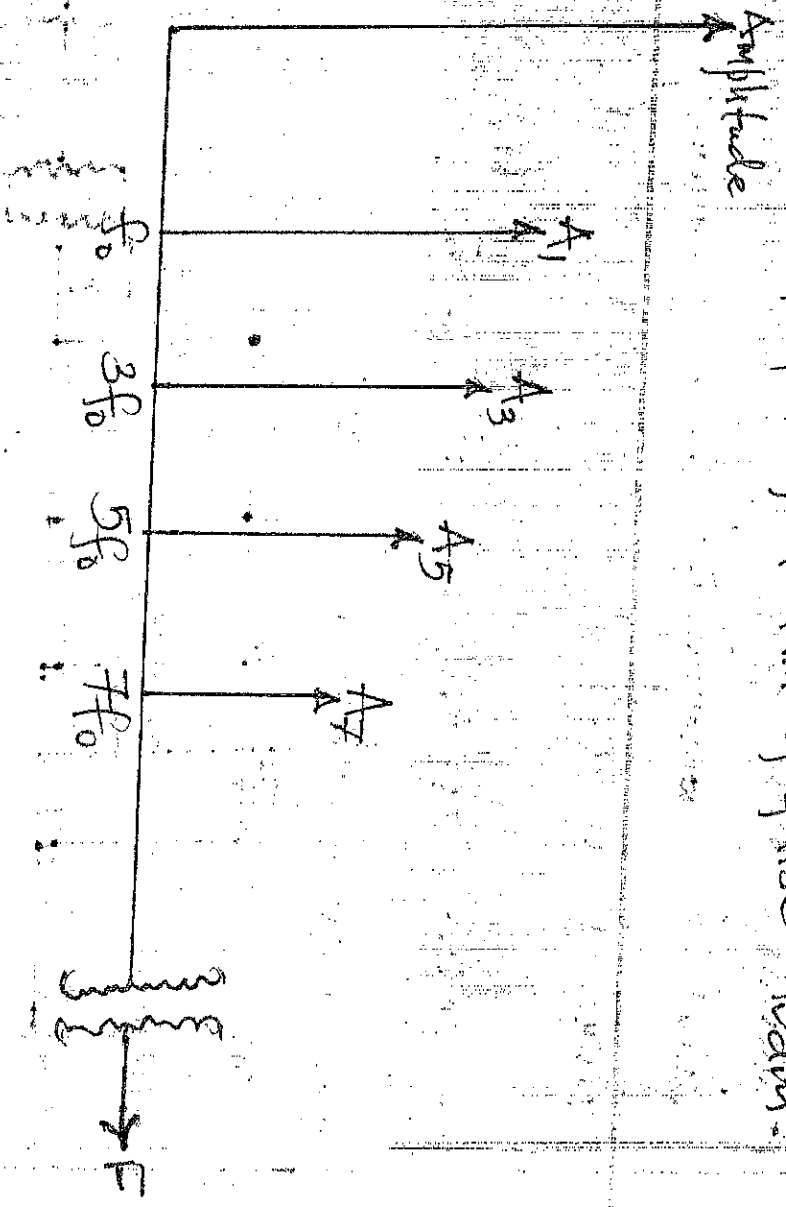
$$b_n = \frac{2}{T_0} \int_{T_0} f(t) \sin(2\pi n f_0 t) dt$$

Therefore;

By integrating over one cycle and substituting into Fourier equation (1) will yield;

$$\Rightarrow f(t) = V + \frac{4 \cdot V}{\pi} \left[\cos \omega t - \frac{1}{3} \cos 3 \omega t + \frac{1}{5} \cos 5 \omega t - \frac{1}{7} \cos 7 \omega t + \dots \right]$$

This implies that pulses signal comprises of many sinusoidal of different frequencies and Amplitude as shown below in frequency domain of pulse train.



PULSE MODULATION

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Pulse Modulation may be defined as a modulation system in which some parameters (i.e. amplitude, frequency or position) of pulse in a pulse train (carrier) are varied in accordance with the instantaneous value of the modulating signal. In this system, waveforms are sampled at regular intervals and the information is transmitted through the sample rate.

In the field of telecommunications, the most widely use of pulse modulation is the Pulse code modulation (PCM).

TYPES OF PULSE MODULATION

The four (4) basic types of pulse modulation techniques are:

1. Pulse Amplitude Modulation

2. Pulse Width Modulation (duration)

3. Pulse Position Modulation

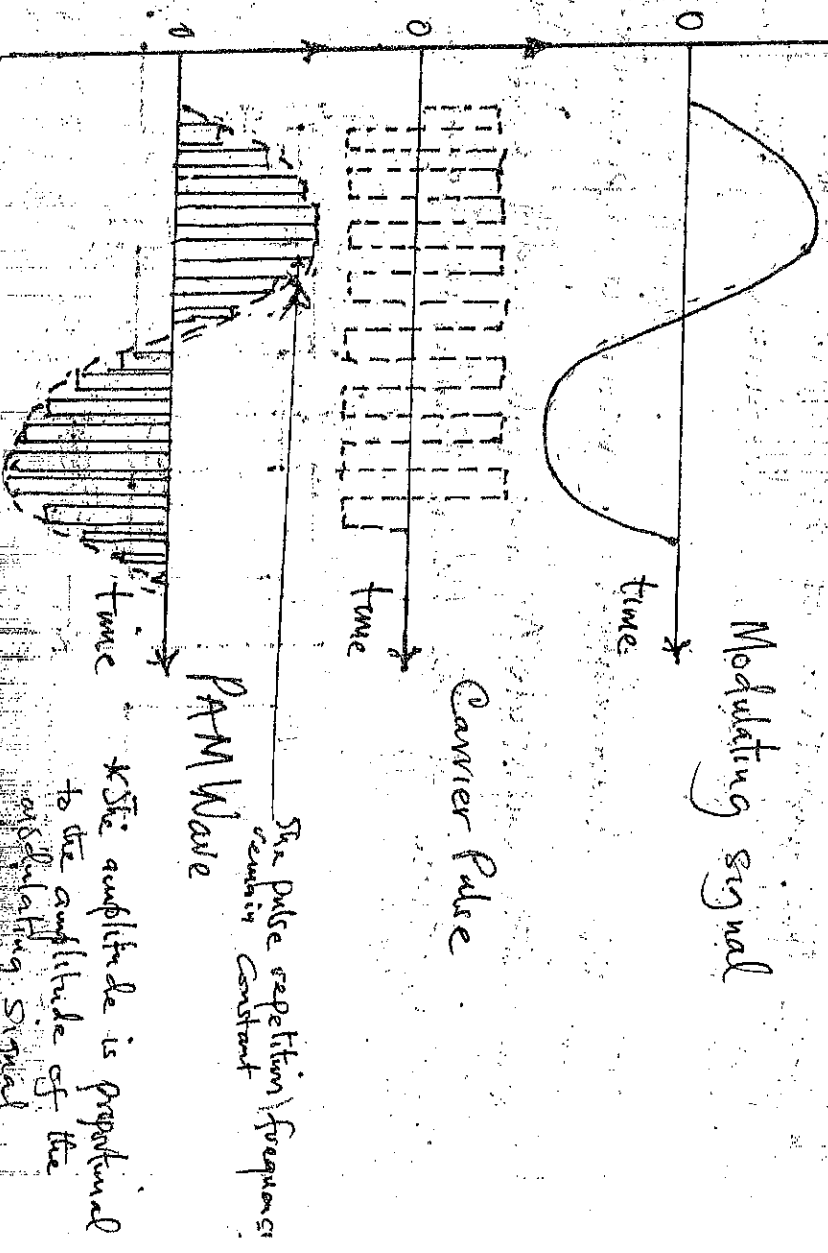
4. Pulse Code Modulation

WHY PULSE MODULATION?

The reason is that, this type of modulation technique allows pulses carrying different modulating signals to be interleaved on time, forming what is known as the "Time division Multiplex (TDM) Signal".

PULSE AMPLITUDE MODULATION (PAM)

This is a form of modulation mode where the amplitude of the carrier pulse train is varied according to the instantaneous value^(carrying msg) of the modulating signal but keeping the width and the position of the pulses constant as given below (Note that the top of the pulse follows the modulating signal).



Advantages of PAM:

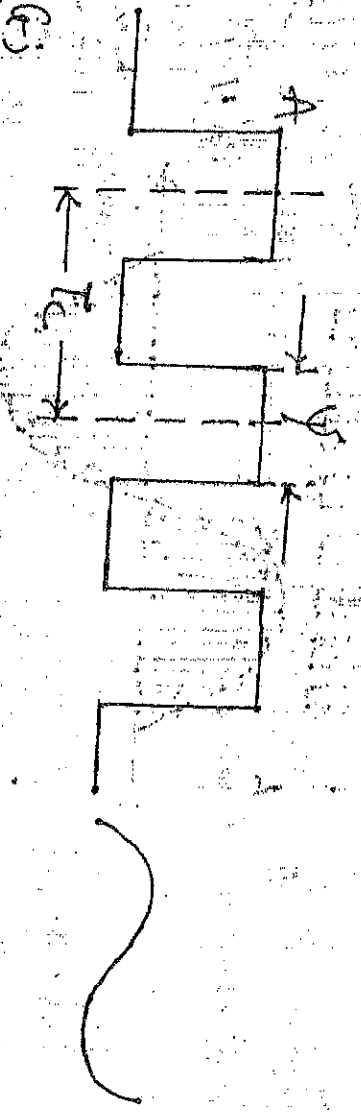
- It uses a very wide bandwidth
- PAM is used to carry information as well as to generate other pulse modulation
- The main advantages of PAM are simple transmitter and receiver designs.

Disadvantages of PAM

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- i.) The Band width required for transmission of a PAM signal is very large as compared to the maximum frequency present in the modulating signal.
- ii.) As the amplitude of the carrier pulses is varied according to the amplitude of the modulating signal, noise increases in the PAM wave.
- iii.) As the amplitude of the PAM signal is varied, it is also varies peak power required by the transmitter.

If the unmodulated carrier is represented by a diagram below with amplitude A and pulse width as τ . Therefore, the periodic time of the pulse train is shown as T_c and τ_a

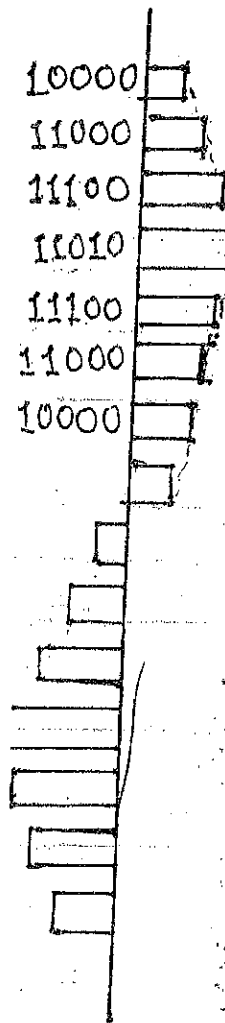


Short-Comings:

- PAM uses very wide bandwidth and so its restricted for transmission purposes.
- = Pulse degradation as the signal propagate may help to distort the intelligent signals.

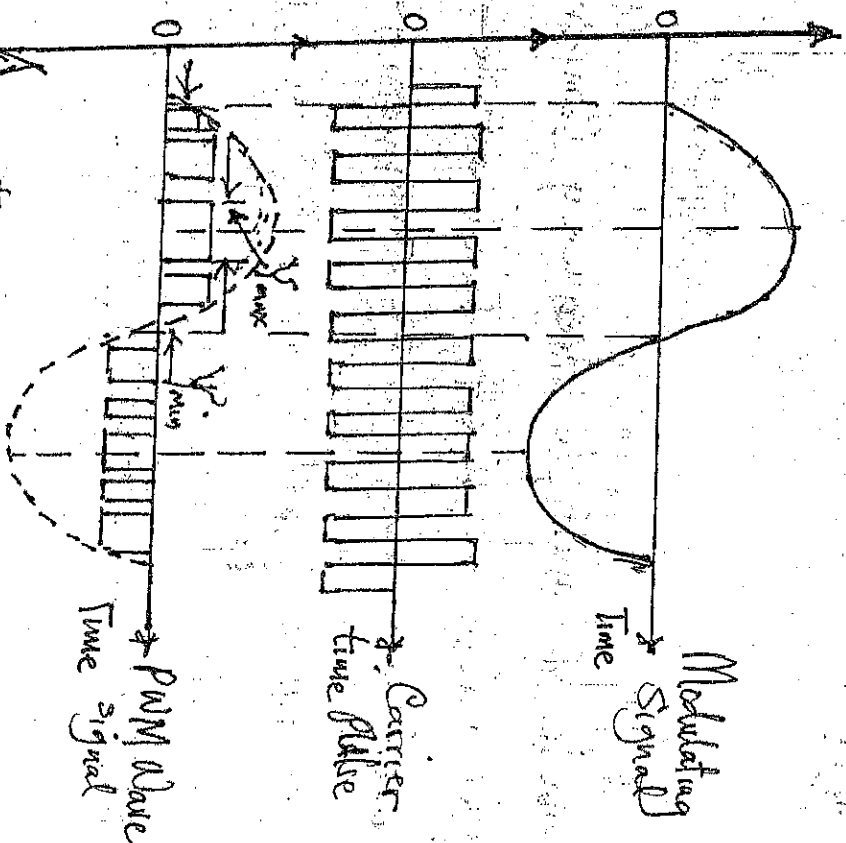
Solution; ~~the~~ the pulses now can be used in ADC to give

Binary result which will be better as shown below



Pulse Modulation (PWM)

This is the modulation where by the amplitude and position of the carrier train is kept constant but the width of the carrier pulse (y) is varied according to the instantaneous value of the modulating signal. This system is also called pulse duration modulation (PDM) or pulse length modulation (PLM). It can be illustrated as below;



Advantages

- i) Noise is less in PWM as the amplitude is kept constant. Note: P_{avg} .
- ii) The signal and noise separation is easy.
- iii) The PWM does not require synchronization b/w transmitter & receiver.

Disadvantages

- i) Large BW is required for PWM communication as compared to PAM.
- ii) The transmitter should be able to handle more power (equivalent to the power of the maximum width pulse).

Mathematically, the width of pulse in PWM signal is given by

$$\Delta_m = A(I + U_m)$$

When there is no message, $U_m = 0$.

Therefore the width of the pulse will be equal to the original width Δ . For positive value of message, the width will be proportional increase by $(1 + U_m)$ factor.

Example for PWM

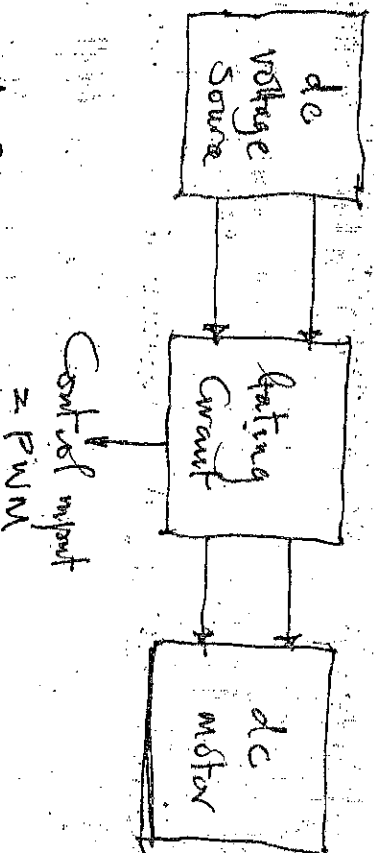
A message signal made of multiple frequency components has a maximum frequency value of 4 kHz . Find out the minimum sampling freq. required according to the sampling theorem.

Given that;

$$F_m = 4\text{ kHz}$$

$$\therefore F_s \geq 2 \times F_m$$

$$= 2 \times 4\text{ kHz} = 8\text{ kHz}$$

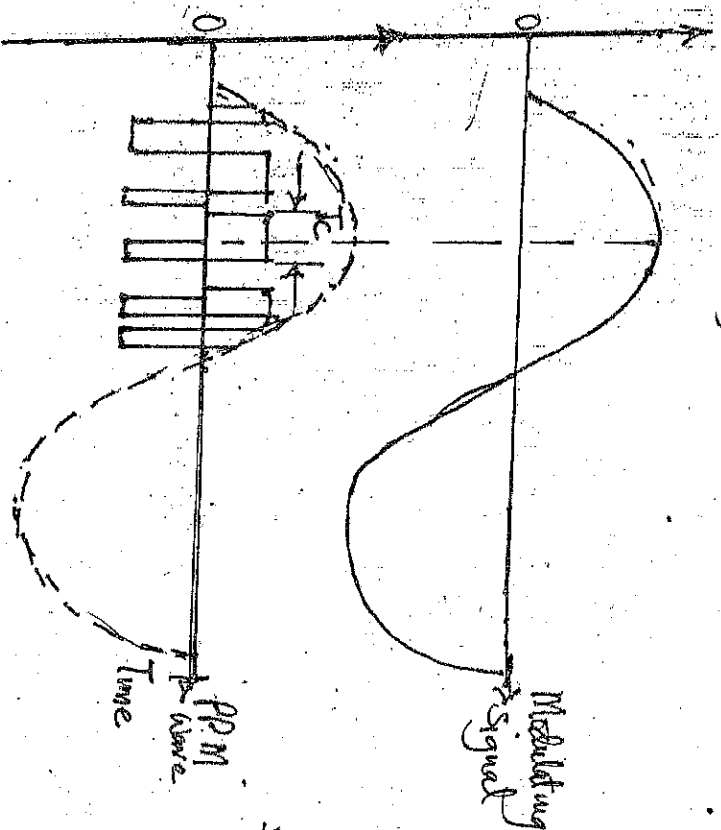


A SIMPLE SPEED CONTROL OF DC MOTOR USING PWM

Pulse Position Modulation (PPM)

22.

In this system, amplitude and width of the carrier pulses are kept constant while the position of each pulse with respect to the position of a reference pulse varied in accordance to the message signal (i.e. the position of the pulse keep on changing with the instantaneous value of the modulating signal for this method, transmitter and Receiver needs to be synchronised as shown below;



* Mathematically the signal is given as $t_p = f(V_m)$

Advantages

- I) As the amplitude and width is kept constant, the transmitter handles constant power because transmitted power already pulse is the same.
- II) It is less noisy due to constant amplitude & the signal to noise separation is easy.

Disadvantages

- I) It needs synchronization between transmitter and receiver.
- II) Large band width is required as compared to PAM.

PULSE CODE MODULATION (PCM)

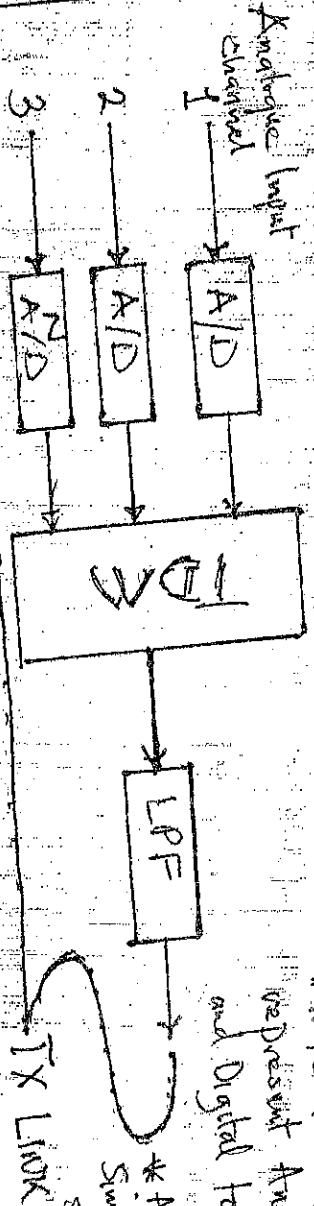
This is a typical form of digital modulation where the analog signal i.e. speech or music must be sampled at regular intervals, using sampling technique but differs as it is digital in nature whereby the samples are then converted into a string of digital codes using a process called "Quantisation".

Quantisation is a process of allocating approximate digital values (numbers) to a series of sampled signals in (Pulse).

PCM therefore modifies the pulses created by PAM to create a completely digital signal; It also quantises the PAM pulses assigning integer values of a specific range to sampled amplitude modulated pulse.

The amplitude of any given pulses is assigned to the nearest level and amplitude variation of a signal can lie anywhere between 0 and 7 volts with respect to some reference level.

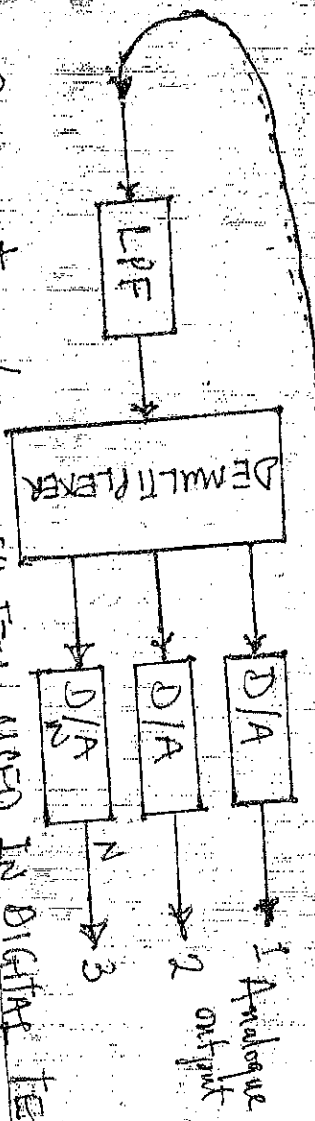
PCM is therefore the most frequently used of all the pulse modulation in digital telephony. The complete system will be drawn as below;



*Note that A/D & D/A.

represent Analogue to Digital and Digital to Analogue in

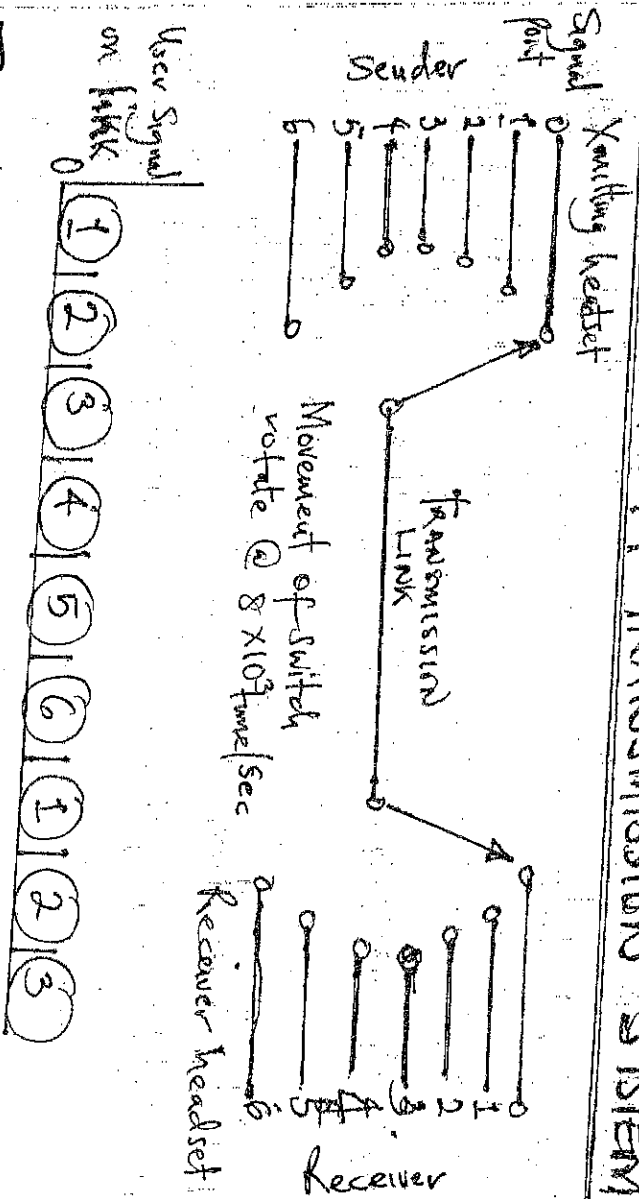
*Also A/D are simply generate signal.



Block Diagram of PCM system used in digital telephony

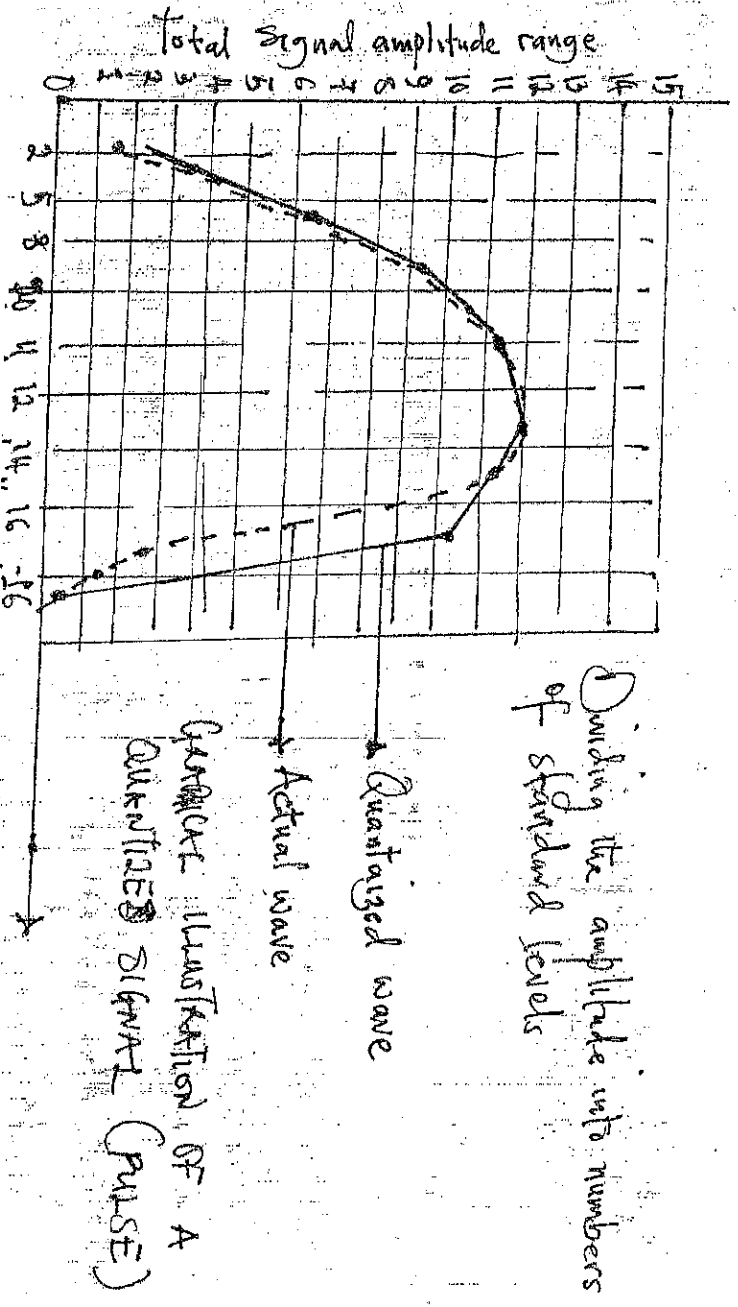
A Simple TDM Transmission System

24



Time Division Multiplex (TDM) is the process whereby each channel is given a time slot (i.e. each sample signal gives rise to a continuous signal after filtering by allowing simultaneous transmission of several signals over a single wide-band link.

The system also scans, transmits, and distributes signals with proper channel synchronization after reconstruction of the signal.



DEMODULATION

25

After modulation is being achieved, the signal is being amplified and propagated using a transmitting antenna; a little of the signal will be obtained from a receiving antenna. At this point, the signal tends to be generally weak, therefore required need for amplification. Probably a received signal might be accompanied by other unwanted signals. The signal now need to be selected and also discarded as to reclaim the original transmitted signal (i.e. the modulated voltages) which are to be recovered. The process of receiving, processing and reconstructing transmitted information can be termed Demodulation or Detection and it is normally done or achieved in the receiver.

A Receiver has the function of selecting a desired signal from all ~~the~~ other unwanted signals, amplifying and demodulating it and finally displaying it in a desirable form. This shows that different modulation techniques requires different receiver types, therefore we have AM, FM, SSB and other receiver types.

The process of recreating original modulating frequency ~~to the original~~ (intelligence) from the rf carrier is referred to as DEMODULATION or DETECTION. The term demodulator is used because the demodulation process is considered to be the opposite of modulation.

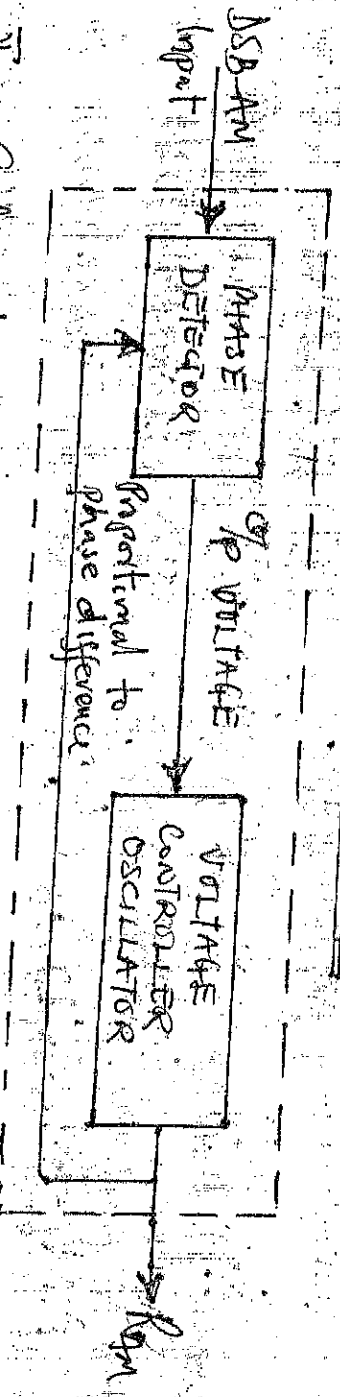
The output of an ideal detector must be an exact reproduction of the modulation existing at the rf wave. Failure to accurately recover this intelligence will result in distortion and degradation of the demodulated signal and intelligence will be lost. The distortion may be in the amplitude, frequency or phase, depending on the nature of the demodulator.

AM DEMODULATION (DSB-AM)

Amplitude modulation refers to any method of modulating an electromagnetic carrier frequency by varying its amplitude in accordance with the message intelligence that is to be transmitted.

The Demodulation of AM-Signal can be broadly classified as Coherent (Synchronous) and non-coherent, where the message can be demodulated.

COHERENT DETECTOR (DSB-AM)



The Coherent detector provides better performance under noisy conditions. Due to the requirement of carrier in the receiver which is in synchronism with that of the transmitter, the coherent detector circuit is more complex compared to non-coherent. In coherent detection, a copy of carrier used for modulation is assumed to be available at the receiver.

Mathematically, the output of the coherent detector can be expressed as:

$$V_{out} = K [I + M \cos(2\pi f_m t)] \cos(2\pi f_c t) \cos(2\pi f_c t)$$

$$K = \int [1 + M \cos(2\pi f_m t)] \cos^2(2\pi f_c t)$$

but

$$\cos^2(2\pi f_c t) = \frac{1}{2} [1 + \cos(4\pi f_c t)]$$

i.e.

$$V_{out} = K [1 + M \cos(2\pi f_m t)] \frac{1}{2} [1 + \cos(4\pi f_c t)]$$

$$= \frac{1}{2} K [1 + M \cos(2\pi f_m t)] + \frac{1}{2} K [1 + M \cos(2\pi f_m t)] \cos(4\pi f_c t)$$

$$= \frac{1}{2} K + \frac{1}{2} K M \cos(2\pi f_m t) + \dots$$

$$= \frac{1}{2} K M \cos(2\pi f_m t) + V_{unwanted}$$

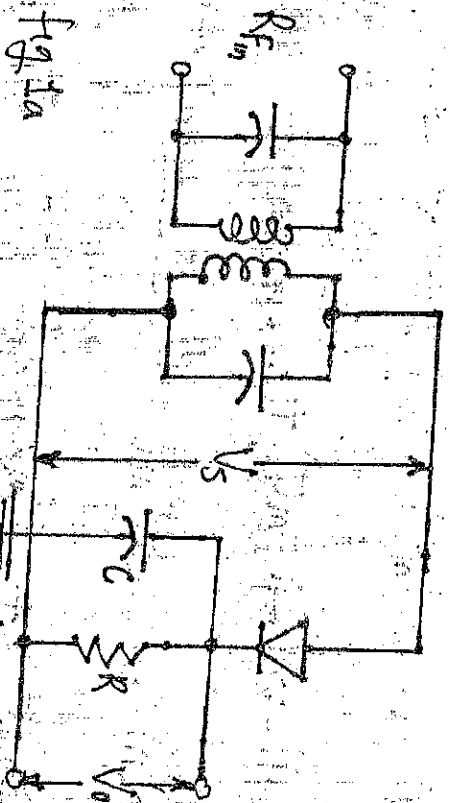
The unwanted terms will be considered as side bands of $2f_c$ and are easily filtered off. The above supply can be put as $(f_c + f_m)$ multiply by f_c which provides with stress as $f_c + f_m$ * $f_c = f_m + 2f_c$

Non-Coherent Detector

A simple non-coherent detector is a semi-conductor diode. The diode is by far the most common device used for AM demodulation (detection).

The circuit below has a combination of a small capacitor C_1 and a large resistance R_1 connected in parallel so as to develop a voltage V_o . From the output V_o at each positive peak of the radio frequency (signal) V_o is $\propto$ V_m .

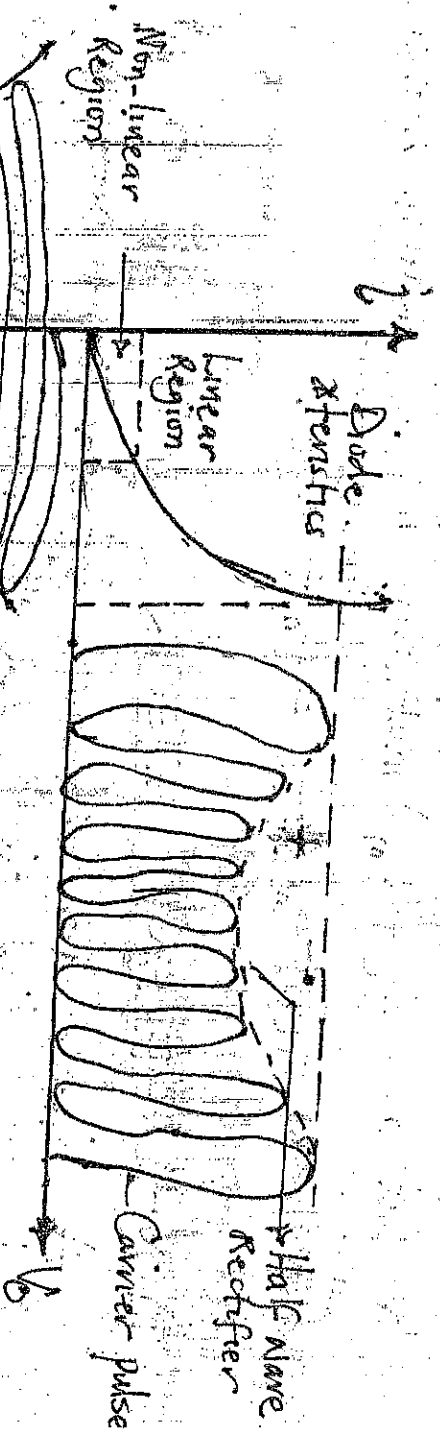
to a potential almost equal to the peak signal voltage V_s . The difference is due to the diode drop since the forward resistance of the diode is small. Between peaks a little of the charge in C decays through R , to be replenish at the next positive peak. The result is the voltage V_o which produces the modulating voltage accurately except for the small amount of RF-ripple.



* In non-coherent detection there is no reference carrier made available at the receiver

A SIMPLE DIODE DETECTOR

The Diode acting as a rectifier as shown below;



DSB-AM
Input signal

Half Wave
Rectifier

Carrier pulse

fig 1b

OPERATION OF A DIODE DETECTOR

⑤ $11.8.8 \text{ freq} = f_c + f_m = 10^6 + 10^3 = 1.001 \times 10^6 \text{ Hz}$
 $1000,000 + 1000 = 1,001,000 \text{ Hz}$
 Using conversion method, $\approx 1,001 \text{ MHz}$

$11.8.8 \text{ freq} = f_c - f_m = 10^6 - 10^3 = 999,000 \text{ Hz} \approx 10^6$
 $\approx 10.999 \text{ MHz}$

⑥ From $V_{m(t)} = A \cos 2\pi f_m t$
 Amplitude of the sidebands is given by $\frac{A}{2}$ and $\frac{C}{2}$
 for the USB and LSB respectively. It can be expressed as:

The USB Amplitude $= \frac{10}{2} = 5 \text{ V}$
 The LSB Amplitude $= \frac{3}{2} = 1.5 \text{ V}$
 or $\frac{\text{Amplitude of USB}}{\text{Amplitude of LSB}} = \frac{5}{1.5} = \frac{10}{3}$

⑦ The fraction of the transmitted power in the side-band
 $P_T = P_C \left(\frac{2 + m^2}{2} \right)$ where P_T = Transmitted Power
 P_C = Power of the Carrier

or $\frac{M^2 P_C}{4} = \text{Power Transmitted in the side-band}$

Where $m = 30$, $P_C = ?$

$P_C \left(\frac{2 + 900}{2} \right) = \frac{2 + 900}{2}$
 ≈ 45.2

$\therefore P_C = 45.2 \text{ W} \approx 45.2 \text{ W}$

What is TDM?

This is an Acronym for Time Division Multiplex. The basic principle is shown below.

Question to Assignment

(Carrier modulation allows the transmission of modulating frequencies without the use of transmission wave as a medium)

$$U_{\text{msg}}: U_c(t) = 10 \cos(2\pi 10^6 t) \text{ V} = f_c = 10^6 \text{ Hz}$$

$$U_m(t) = 3 \cos(2\pi 10^3 t) \text{ V} = f_m = 10^3 \text{ Hz}$$

From General Equation $\Rightarrow U_c(t) = A \cos(2\pi f_c t + \phi) = f_c = 10$
 $U_m(t) = a \cos(2\pi f_m t)$

For modulated signal, New Carrier amplitude (A) is given as
 $A = K + U_m(t)$

Therefore, Computing the two expressions, $\rightarrow f_m = 10^3 \text{ Hz}$

$$U_c(t) = 10 \cos(2\pi 10^6 t) \quad \text{i.e. } U_c(t) = A \cos(2\pi f_c t)$$

Given that modulating signal amplitude divided by Unmodulated Carrier Amplitude
 $M = \frac{a}{K} = \frac{3}{10} = 0.3$
 $m = ?$

Depth of Modulation (or modulation index) $= \frac{3}{10} \times 100$

$$\textcircled{b} \cos A \cos B = \frac{1}{2} [\cos(A-B) + \cos(A+B)]$$

$$= \frac{1}{2} \cos(2\pi 10^4 t) \cos(2\pi 10^3 t) = \frac{1}{2} \left[\cos(2\pi 10^4 + 2\pi 10^3 t) + \cos(2\pi 10^4 + 2\pi 10^3 t) \right]$$

$$= \frac{1}{2} \cos(2\pi 10^4 - 2\pi 10^3 t) + \frac{1}{2} \cos(2\pi 10^4 + 2\pi 10^3 t)$$

$$\frac{1}{2} \cos 2\pi (f_c - f_m) t + \frac{1}{2} \cos 2\pi (f_c + f_m) t$$

AM DEMODULATION (DIODE DETECTOR) (27)

The Maximum value of an applied modulation index which a diode detector will handle without negative peak clipping can be calculated as;

The modulation index in the demodulated wave will be

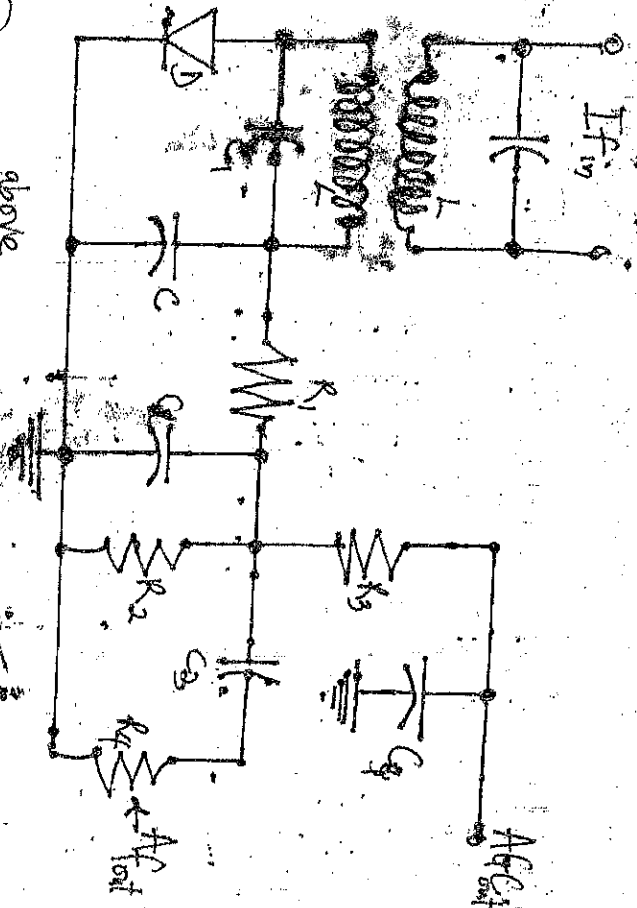
given as;

$$M_d = \frac{I_m}{I_c} = \frac{V_m/Z_m}{V_c/R_c} = M \frac{R_c}{Z_m}$$

Since the Maximum tolerable modulation index in the diode output is unity, the Maximum permissible transmitted modulation index will be given as follows;

$$M_{\max} = M_{d\max} \frac{Z_m}{R_c} = 1 \frac{Z_m}{R_c} = \frac{Z_m}{R_c}$$

Problem 1;



From the above circuit let the various resistances & $R_1 = 110 \text{ k}\Omega$, $R_2 = 220 \text{ k}\Omega$, $R_3 = 470 \text{ k}\Omega$ and $R_L = 1 \text{ M}\Omega$. What is Maximum modulation index to be applied to this diode set without causing any negative peak clipping.

Given that;

$$R_c = R_1 + R_2 = 110 + 220 \\ = 330K$$

$$Z_m = \frac{R_2 R_3 R_4}{R_2 R_3 + R_3 R_4 + R_4 R_2} + R_1$$

$$= \frac{220 \times 470 \times 1000}{220 \times 470 + 1000 + 1000 \times 220} + 110 = 130 + 110$$

$$Z_m = 240K$$

$$\therefore M_{\max} = \frac{Z_m}{R_c} = \frac{240}{330} = 0.73 \approx 73\%$$

2x2) As Assignment

* What is the Maximum bandwidth of an fm signal with a deviation of 30KHz and a maximum modulating signal of 5KHz as determined (a) Modulation index (b) Carson's rule.

Ans: The bandwidth of an fm signal can be calculated by using the modulation index and by Carson's rule.

2x2) The AM Broadcast band consists of 107 channels for stations 10KHz wide. The maximum permitted modulation freq is 5KHz, could fm be used on this band? If so explain what would be necessary to make it happen.

3) Explain the operation of a balance slope detector using a circuit diagram and its response characteristics, and also in what ways is this ckt an improvement on the slope detector and in turn, what are its disadvantages?

FM DETECTORS (DEMODULATION)

(31)

This is the process by which ~~the~~ intelligent signal is retrieved from the frequency modulating signal. FM detector has an advantage of being immune to noise.

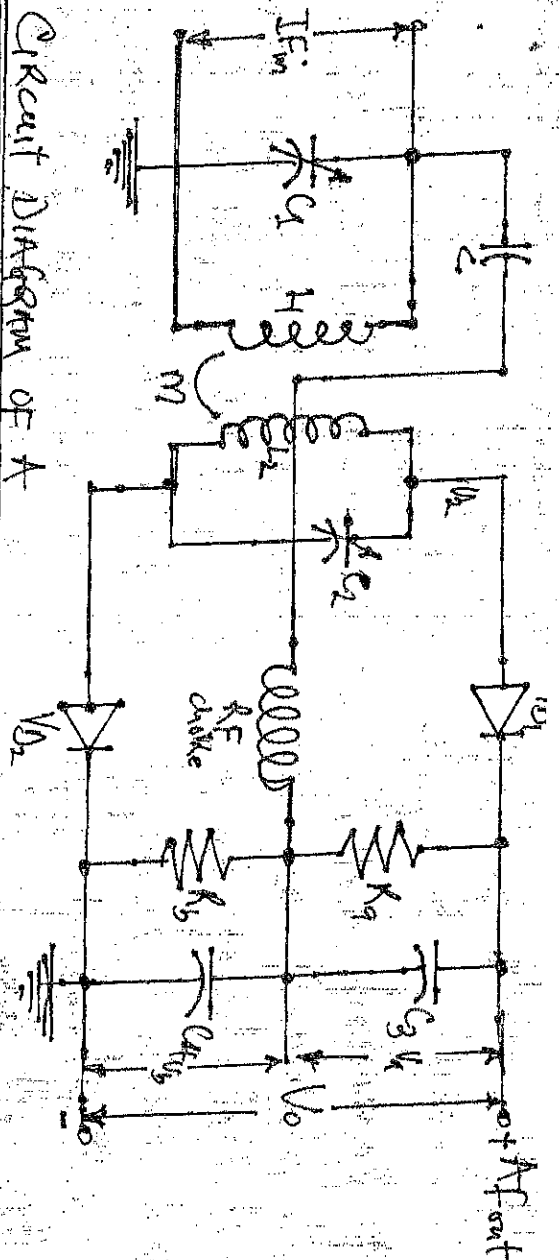
Any Ckt that will convert a freq. variation in the carrier back to a proportional voltage variation can be used to demodulate or detect fm signals. Meaning that, a circuit used in recovering the original modulating signal from an fm transmission are called demodulators, detectors or discriminators.

TYPES OF FM DETECTOR INCLUDE;

- Phase Discriminators
- Ratio Detector
- Slope Detector
- Balanced-slope-Detector (Also called Trancie detector).

Phase Discriminator;

This is also known as the Center-tuned discriminator or the Foster-seely discriminator or detector. This is a circuit where all ^{other} fm detector which detect fm signal by giving an output voltage whose amplitude is proportional to the freq. of the input signal. This detector is widely used in practice especially in fm radio receivers (wide band or narrow band) and also in Satellite Station receiver for reception of TV carrier. It's preferred in situations in which linearity is important. (e.g. high quantity fm receivers).



Circuit Diagram of A
Foster-Seeley Discriminator

PRINCIPLE OF OPERATION:

This circuit utilizes the effect of phase shift between the primary and secondary voltage of a tuned transformer which are both tuned to the carrier frequency of the incoming signal. The phase shift is frequency dependent and so the modulating signal is recovered. This is desirable because it greatly simplifies alignment and also base on the fact that the process yields better linearity. The output is taken across the R-C assembly, ~~V~~ the voltage ~~applied~~ to the each diode is the sum of the primary voltage and the corresponding half secondary voltage.

Note that the phase shift increases as the frequency decreases and decreases as the frequency increases. This current must be fed from amplitude variation for proper operation. It will also show that ~~the~~ the primary and secondary voltages are given as:

- 1- ~~2x~~ 90° out of phase when the input freq is f_c
- 2- Less than 90° out of phase when f_{in} is higher than f_c
- 3- More than 90° out of phase when f_{in} is below f_c .